



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

Assignment 3

Programme	: Master of Cyber Security
Subject Code	: MECR1063
Subject Name	: Cryptographic Engineering
Session-Sem	: 2024/2025-1

Prepared by : 1) Keertennah Devi A/P Ponnambalam (MEC254026)

Section : 01 / 52 (ODL)

**Group /
Member ID** : 2A

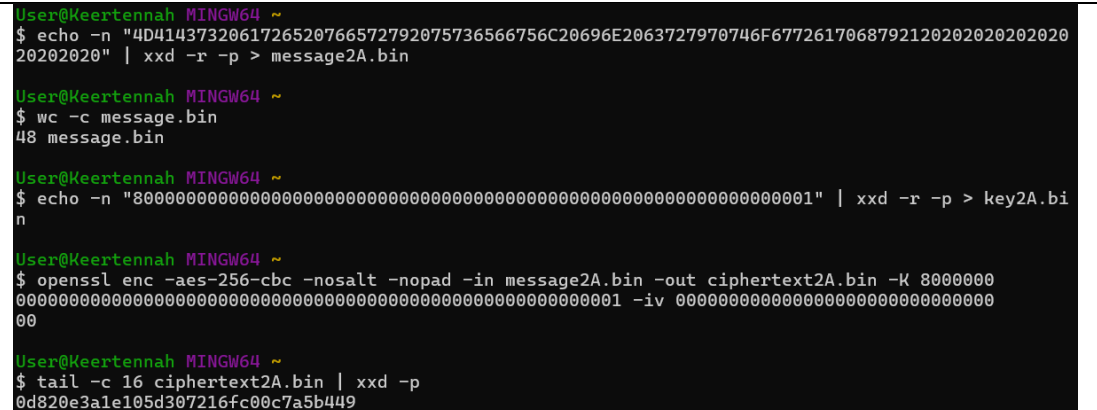
Lecturer : Dr. Muhalim Bin Mohamed Amin

Video Link :

Date : 23 June 2026

Turnitin (%)		Marks / Remarks
Similarity	AI	

Lab 2: Integrity Mechanisms: Hashing / MAC using Library Course: MECR1063 Cryptography Engineering (2025/2026-2)			
Name:	KEERTENNAH DEVI A/P PONNAMBALAM	Matric:	MEC254026
Member ID:	2A	Section:	52

PART A: Generating CBC-MAC of a message	
A 1	Command: <pre>echo -n "4D4143732061726520766572792075736566756C20696E2063727970746F 6772617068792120202020202020202020" xxd -r -p > message2A.bin</pre>
A 2	Command: <pre>echo -n "8000 0001" xxd -r -p > key2A.bin</pre>
A 3	Command: <pre>openssl enc -aes-256-cbc -nosalt -nopad -in message2A.bin -out ciphertext2A.bin - K 8001 -iv 00000000000000000000000000000000</pre>
A 4	Command: <pre>tail -c 16 ciphertext2A.bin xxd -p</pre>
A 5	Output: <pre>0d820e3a1e105d307216fc00c7a5b449</pre>
A 6	Screenshot the commands consisting of the steps of A2 – A4:  <pre>User@Keertennah MINGW64 ~ \$ echo -n "4D4143732061726520766572792075736566756C20696E2063727970746F6772617068792120202020202020202020" xxd -r -p > message2A.bin User@Keertennah MINGW64 ~ \$ wc -c message.bin 48 message.bin User@Keertennah MINGW64 ~ \$ echo -n "8001" xxd -r -p > key2A.bi n User@Keertennah MINGW64 ~ \$ openssl enc -aes-256-cbc -nosalt -nopad -in message2A.bin -out ciphertext2A.bin -K 80000000 0001 -iv 00000000000000000000000000000000 00 User@Keertennah MINGW64 ~ \$ tail -c 16 ciphertext2A.bin xxd -p 0d820e3a1e105d307216fc00c7a5b449</pre>

[illegible]

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